

Tree Ensemble Modeling for Heart Disease Classification

Austin Irwin, Shane Faberman, Rhett Lavender, Al Pagar

1 Problem Description

In this project, our team aimed to develop a machine learning model to classify whether or not a patient has heart disease. By accurately predicting whether or not a patient has heart disease, physicians would be able to be more sure of their diagnosis and provide better patient care.

Heart disease is the leading cause of death in the United States [2]. In addition to the hundreds of thousands of known deaths, thousands of Americans die each year from misdiagnosed heart attacks [7]. The Centers for Disease Control and Prevention estimates that over \$400 billion is lost annually from heart disease, including the cost of healthcare services, medicine, and lost productivity from those affected by heart disease.

We plan on using a Kaggle data set called Heart Disease Classification [1], which gives various features and a binary target feature that determines whether or not a patient has heart disease. Our goal was to train a machine learning model to predict whether a patient has heart disease, given those features.

A preliminary description of some of the features in the data set can be seen below:

Feature	Description
age	Age of the person.
sex	Gender of the person.
chest_pain_type	Typical angina, atypical angina, non-anginal pain, or asymptomatic
resting_bp	Blood pressure while resting (in mm Hg on admission to the hospital).
cholesterol	Serum cholesterol in mg/dl.
fasting_blood_sugar	Blood sugar while fasting.
restecg	ECG (electrocardiographic) while resting
max_hr	Maximum heart rate achieved.
exang	Exercise-induced angina.
oldpeak	ST depression induced by exercise relative to rest.
slope	The slope of the peak exercise ST segment
num_major_vessels	Number of major vessels colored by fluoroscopy.
thal	Thalassemia
target	Whether or not the patient has the disease.

Table 1: Description of dataset features.

2 Related Works

Due to the large number of deaths from heart disease every year, the prediction of heart disease has been an important problem to solve, particularly in the field of machine learning. In one study, a CART decision tree was used for the

classification of heart disease. They found that a patient's ST Slope, the rate of change in the ST segment relative to an exercise-induced increase in heart rate, was the most important feature in their model. It had a characteristic importance of 28%, which was 10% higher than the second most important characteristic [6].

Another study carried out by students and faculty of the University of Galati tested seven different binary classification machine learning models on the same heart disease classification data set. Random forests, logistic regression, naive Bayes, k-nearest neighbors (kNN), support vector machines (SVM), gradient boosting, and artificial neural networks. In their study, the hyperparameters were tuned using grid-search cross-validation. The kNN model had the highest accuracy (99%) and highest AUC (0.99) [3].

A study published in July 2024 used a stacked ensemble method with Naive Bayes, logistic regression, and SVM models to provide more accurate and holistic predictions. In addition, the firefly algorithm was used to optimize the feature selection process. Using the stacking ensemble method on the substantial features chosen by the firefly algorithm gave a maximum accuracy of 86.79% [5].

3 Data Description

To gain a better understanding of the dataset, we visualize the distributions of its individual features alongside the distribution of the target variable. This helps reveal underlying patterns, identify potential outliers, and assess whether any variables exhibit irregularities that may influence the training of the models.

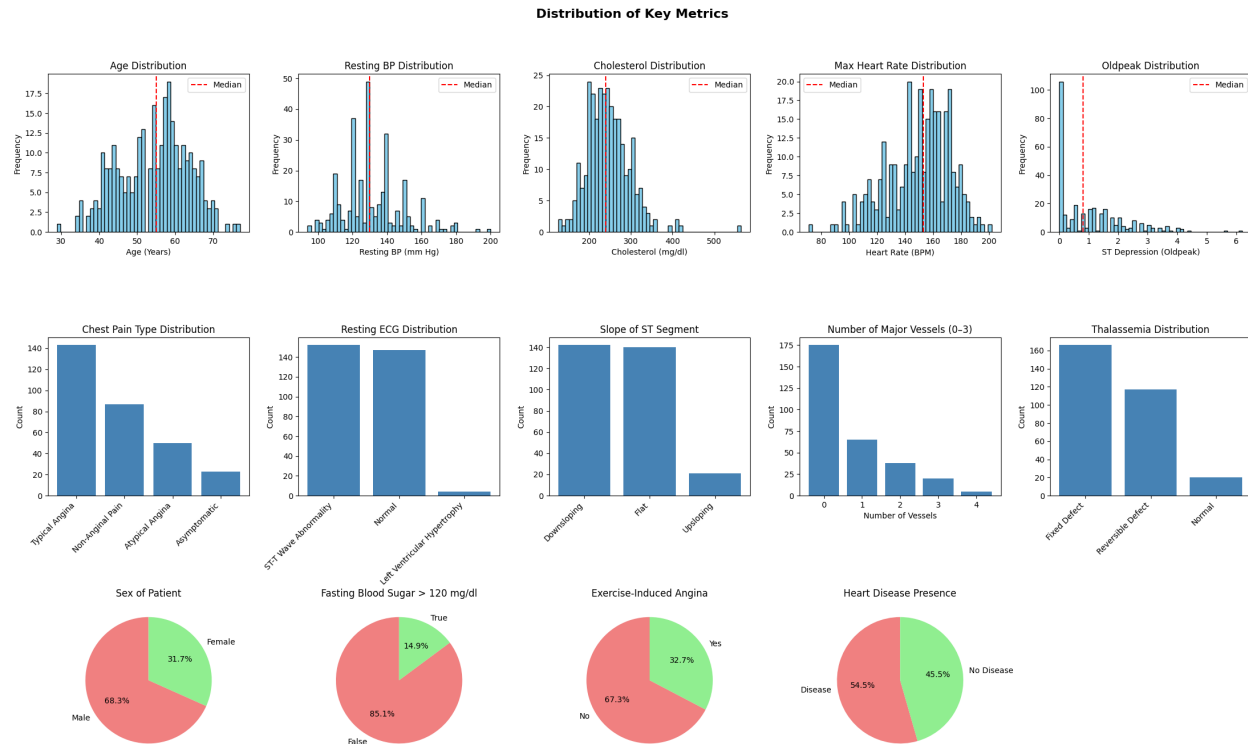


Figure 1: Key features and target variable distributions

The histograms across the top of Figure 1 show that these continuous features have clear, interpretable distributions, although a few, like cholesterol and ST depression, are somewhat skewed. The bar charts and pie charts along the bottom show similar patterns for the categorical features. For example, about two-thirds of patients are male, most have normal fasting blood sugar, and most do not experience exercise-induced angina.

Finally, the pie chart on the right shows that the target variable, heart disease presence, is roughly balanced between positive and negative cases.

Overall, the dataset required very little cleaning or feature engineering before training our models.

4 Methods

For our analysis, we looked at AdaBoost, Gradient Boosting, and Random Forest models. These models were built using scikit-learn functions. We split the data into 80% train and 20% test sets, and we used the GridSearchCV function to tune the hyperparameters for each model. This was done to prevent overfitting and produce more reliable, generalizable models. We then compared the models by looking at metrics such as accuracy, ROC-AUC Score, and False Negative Rate evaluated on the test set.

5 Results

5.1 AdaBoost

Adaptive boosting, or AdaBoost, is an ensemble learning technique that combines many weak learners to create a strong classifier. Initially, equal weights are assigned to all training samples. As weak learners are trained iteratively, these weights are adjusted so that misclassified data points carry greater weight and contribute more to model error. By combining multiple weak learners, the AdaBoost algorithm enhances predictive accuracy for both classification and regression tasks. The algorithm is easily implemented in the Python programming language using the scikit-learn package. An AdaBoostClassifier object is defined along with a parameter grid for tuning, as seen below:

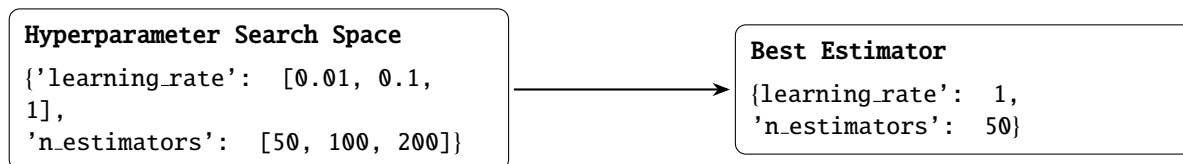


Figure 2: Hyperparameter search space and selected best estimator from model tuning for AdaBoost Model.

Using scikit-learn's GridSearchCV function, hyperparameter tuning was performed, yielding a best estimator composed of 50 depth-1 decision trees (stumps) and with a learning rate of 1. Following tuning, the best estimator was thoroughly evaluated on the testing set. The best AdaBoost model achieved an accuracy of roughly 82 percent and a ROC-AUC score of about 0.915. The model relied heavily on three features, in particular: ST depression induced by exercise relative to rest (*oldpeak*), cholesterol, and the number of major vessels colored (*num_major_vessels*). In contrast, a patient's blood sugar while fasting and electrocardiographic (ECG) while resting had close to no influence on the model's predictions. The confusion matrix, the ROC curve, and the feature importance plot for the best AdaBoost model can be seen below. Based on the confusion matrix, it is evident that this particular model has a fairly high false negative rate, which is typically a metric that we seek to minimize over other metrics. Ideally, we will not be moving forward with the AdaBoost model.

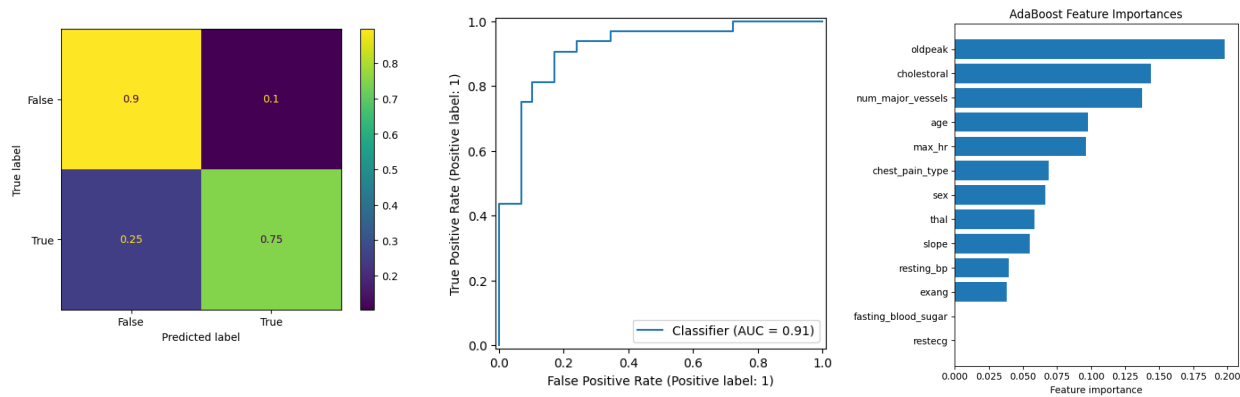


Figure 3: AdaBoost Model evaluation plots

5.2 Gradient Boosting

The next model we considered was a gradient boosting model, using scikit-learn's GradientBoostingClassifier. Gradient boosting is an ensemble learning technique that aggregates Decision Trees by fitting weak learners to the pseudo-residuals of the previous tree. These weak learners have high bias, but low variance, and the aggregation of them reduces the bias for the overall model. The parameter grid was defined as follows:

Hyperparameter Search Space

```
{'max_depth': [2, 3, 4, 5,
None],
'min_samples_leaf': [1, 2, 3],
'min_samples_split': [2, 3, 4],
'learning_rate': [0.05, 0.1,
0.2],
'n_estimators': [75, 100, 125,
150]}
```

Best Estimator

```
{'max_depth': 2,
'min_samples_leaf': 1,
'min_samples_split': 4,
'learning_rate': 0.1,
'n_estimators': 125}
```

Figure 4: Hyperparameter search space and selected best estimator from model tuning for Gradient Boosting Model.

The full training set was trained on the best estimator: 'learning_rate': 0.1, 'max_depth': 2, 'min_samples_leaf': 1, 'n_estimators': 125, 'min_samples_split': 4. The model achieved 83.6% accuracy and a 0.900 ROC-AUC Score on the test set and produced the following confusion matrix and ROC Curve. The number of major vessels colored was the most important feature in the gradient boosting model, per scikit-learn's feature importance method. Similarly to AdaBoost, a patient's fasting blood sugar and resting electrocardiographic values had almost no effect on the model.

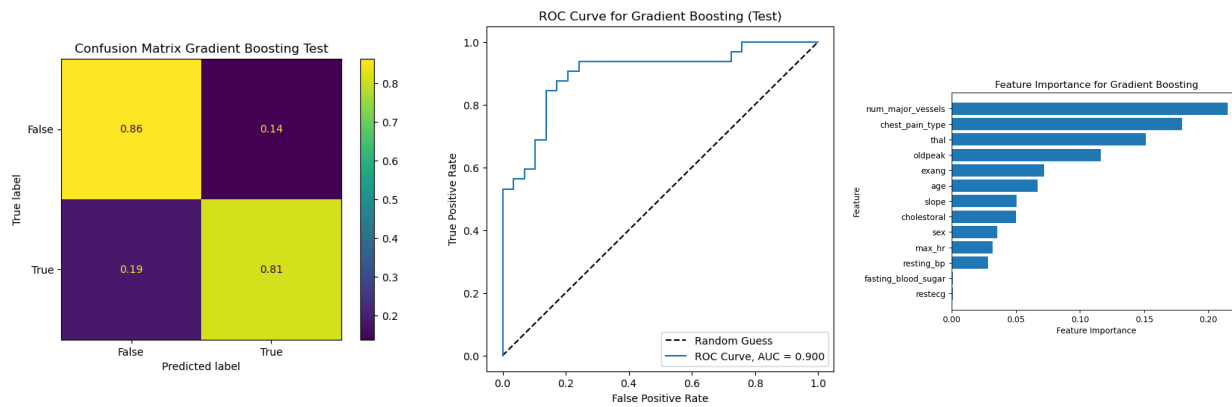


Figure 5: Gradient Boosting Model evaluation plots

5.3 Random Forest

Random Forests are an ensemble learning technique that build multiple Decision Trees using two layers of randomness: bootstrap sampling of rows and random feature selection at each split. By averaging the predictions of many decorrelated trees, Random Forests drastically reduce the variance present in a single large Decision Tree.

We trained a RandomForestClassifier and used GridSearchCV to tune five key hyperparameters, including tree depth, number of trees, minimum samples per split and leaf, and the number of features considered at each split.

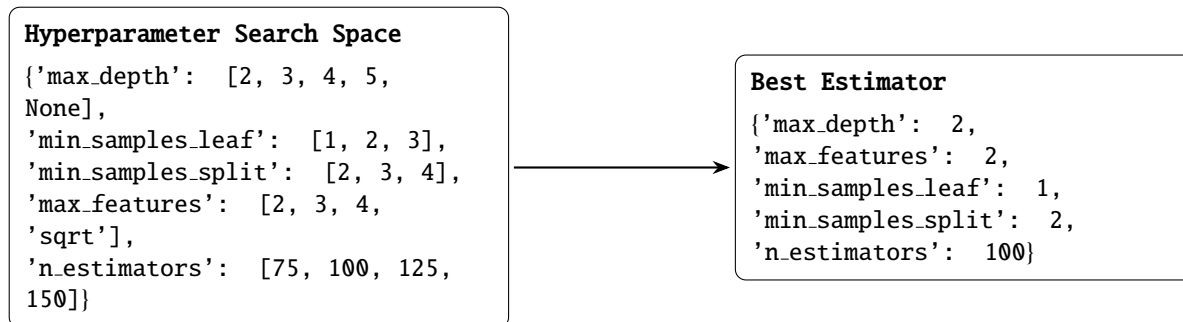


Figure 6: Hyperparameter search space and selected best estimator from model tuning for Random Forest Model.

The best-performing model, shown in Figure 8, uses shallow trees with a max depth of 2 and 100 estimators.

On the test set, this model achieved a test accuracy of 87 percent, and the ROC curve in the center of Figure 8 shows a strong ROC-AUC of 0.94, indicating very good discrimination between patients with and without heart disease. The confusion matrix on the left of Figure 8 shows relatively low misclassification rates for both classes. Finally, the feature importance plot on the right of figure 8 suggests that thalassemia, chest pain type, ST depression, and exercise-induced angina are among the most influential predictors in the model.

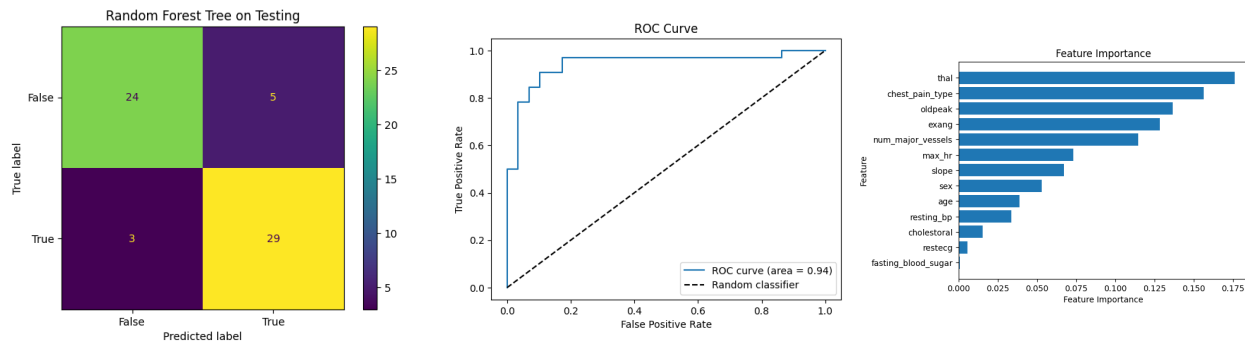


Figure 7: Random Forest Model evaluation plots

Calculating the permutation importance of this Random Forest Model we found that 5 features had negative importance. That means the model actually improved when that variable was permuted randomly. These features were age, max_hr (maximum heart rate achieved), resting_bp (blood pressure while resting), cholesterol, and oldpeak (ST Depression induced by exercise relative to rest). To further explore this we removed these 5 features from the dataset and retrained the model. The accuracy improved from 87 percent to 90 percent, although the ROC-AUC Score dropped marginally from 0.94 to 0.93

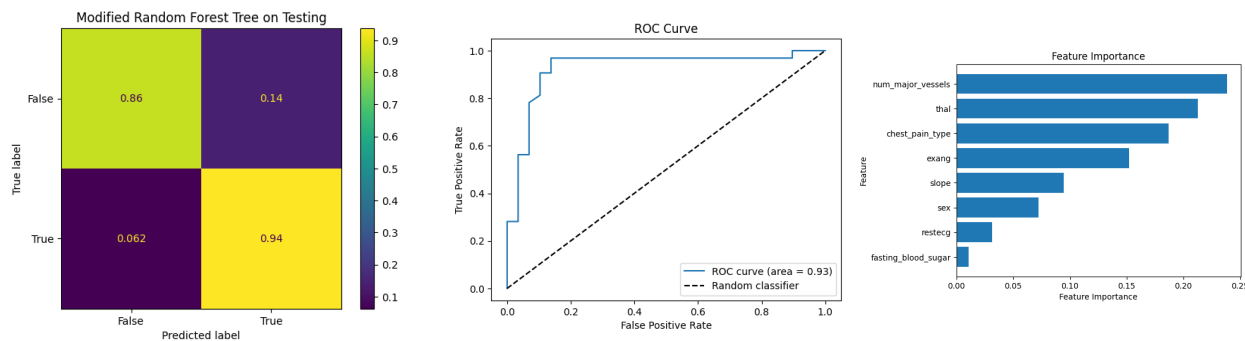


Figure 8: Modified Random Forest Model evaluation plots

6 Conclusion

Out of all three tree ensemble methods tested, the Random Forest technique clearly performed the best on the testing data, with an accuracy of approximately 87-percent, and a ROC-AUC score of 0.94. This particular level of performance in classifying heart disease suggests the model may provide the most value as a decision-support tool. In other words, the prediction of the final Random Forest model is **not** a substitute for a professional diagnosis. The model's accuracy is simply not high enough to warrant relying solely on its predictions to make a definitive diagnosis.

Our best model depended most on whether or not the patient has thalassemia and the type of chest pain the patient is experiencing. These predictors are a matter of symptomatic presentation. If doctors and nurses can improve accuracy of feature measurement, risk-screening tools such as our Random Forest model may subsequently improve.

7 Future Directions

One issue with our study is the lack of data. The data set we used from Kaggle only contained 303 rows, far too few for us to consider our results significant. As undergraduate students, we are limited to publicly available data, and

health data is rightfully protected and out of the public's reach. To continue our study, we would need to be authorized to obtain and use private medical data.

A related issue is that due to the limited availability of data on this topic, there was no feature related to race. Studies have suggested that race plays a role in heart disease, as certain races are disadvantaged regarding heart disease treatment [4]. In the future, we'd like to consider race in our model to see how error rates differ by race.

Finally, our model takes in symptomatic data and predicts heart disease. The July 2024 study using the stacked ensemble method mentioned previously uses similar symptom-based data and achieves a similar maximum accuracy. We would like to consider genetic history, diet, and exercise data and see if we can achieve a better model. Our goal is heart disease prevention, so flagging patients as at-risk before they show symptoms will help prevent future heart disease cases and deaths.

References

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